Homework 2: Elliptic Curve Cryptography (Toy Model) and Caesar Cipher

Cryptography and Security Course

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1 Introduction and ECC Cryptographic Scheme (Toy Model)

In the first part of this work, we extend the toy model based on the *secp*256k1 elliptic curve reduced modulo $p = 17$, defined by the equation:

$$y^2 \equiv x^3 + 7 \pmod{17}$$

with base point $G = (1, 5)$ and group order $N = 16$ (including the 15 affine points and the point at infinity)[cite: 1].

1.1 Phases of the ECC Encryption/Decryption Algorithm (ElGamal Style)

Using the asymmetric encryption scheme based on elliptic curves:

1. **Key Generation:** The user chooses a private key $d \in [1, 15]$ and computes the public key as $Q = d \cdot G$.
2. **Encryption:** Given a message M encoded as a valid point on the elliptic curve:
   - A random ephemeral integer $r \in [1, 15]$ is chosen.
   - The first component of the ciphertext is computed: $C_1 = r \cdot G$.
   - The second component of the ciphertext is computed: $C_2 = M + r \cdot Q$.
   - The ciphertext is the pair of points (C_1, C_2).
3. **Decryption:** Upon receiving (C_1, C_2), the recipient possessing the private key d recovers the message by subtracting $d \cdot C_1$ from point C_2:

$$M = C_2 - d \cdot C_1 = (M + r \cdot Q) - d(r \cdot G) = M + r(d \cdot G) - r(d \cdot G) = M$$

1.2 Role of the Keys

- **Private Key (d):** Maintained strictly secret, it is required to decrypt messages and sign transactions.
- **Public Key (Q):** Derived from $d \cdot G$, it is shared publicly to allow anyone to encrypt a message intended for the owner.
- **Ephemeral Key (r):** A random value chosen for each individual encryption operation; it guarantees *semantic security* (encrypting the same message twice produces distinct ciphertexts).

2 Implementation of the Caesar Cipher

The Caesar cipher is a monoalphabetic substitution cipher in which each letter in the plaintext is shifted by a fixed number of positions k (the key) in the alphabet. Formally, for encryption:

$$C = (P + k) \pmod{26}$$

And for decryption:

$$P = (C - k) \pmod{26}$$

3 Comparative Analysis Between the Two Schemes

Feature	ECC Scheme (Toy)	Caesar Cipher
Encryption Type	Asymmetric (Public/Private Key)	Symmetric (Shared Secret Key)
Key Space	Reduced (15 valid values)	Very small (26 possible values)
Mathematical Operations	Modular elliptic curve arithmetic	Basic modular arithmetic modulo 26
Statistical Properties	Complete masking via the ephemeral key r	Statistical invariance (preserves language letter frequencies)

Table 1: Comparison between the ECC Toy Scheme and the Caesar Cipher

4 Statistical Cryptanalysis of the Caesar Cipher

Since the Caesar cipher does not mask the frequency distribution of the language, it can easily be broken using statistical analysis based on the **Chi-squared (χ^2) test**.

4.1 Key Recovery Algorithm Design

The algorithm tests each of the 26 possible shifts k:

1. Decrypts the ciphertext tentatively using the candidate key k.
2. Calculates the observed frequency O_c of the letters in the decrypted text.
3. Compares the observed distribution against the expected frequencies E_c of the target language (e.g., 'A' $\approx 11.7\%$, 'E' $\approx 11.8\%$, etc.) using the metric:

$$\chi^2 = \sum_{c \in \text{alphabet}} \frac{(O_c - E_c)^2}{E_c}$$

4. Selects as the correct key the one that minimizes the χ^2 value.

4.2 Testing on Ciphertexts of Varying Lengths and Effectiveness

- **Long Ciphertexts (> 200 characters):** The algorithm identifies the correct key with near-perfect accuracy because the law of large numbers guarantees that letter frequencies in the text closely align with standard linguistic statistical distributions.

- **Short Ciphertexts (< 30 characters):** The effectiveness decreases noticeably. Due to sample variance, a short text may exhibit a severely distorted letter distribution (e.g., absence of the letter 'Q' or overrepresentation of 'Z'), leading to false positives in the Chi-squared test.